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Comments:			
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Flight Program:	ISS		
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gangopadhyay, Anup Ph.D. (Washington University in St. Louis)		
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	This grant is a continuation of previous grants NNX10AU19G and NNX16AB52G. Funding is provided for Prof. Dr. Kenneth Kelton and Dr. Anup Gangopadhyay for their role in incorporating all ground-based studies and flight preparation activities by the QUASI team at Washington University in St. Louis (WUSTL). These support QUASI flight investigations on the International Space Station (ISS) and collaborative flight projects with ESA teams ICOPROSOL (scientific coordinator – Dr. Dirk Holland-Moritz, DLR) and THERMOPROP (scientific coordinator – Prof. Dr. Hans Fecht, Ulm University). The flight experiments on the ISS are in designated batches. Batch 1 and batch 2 experiments have now been successfully completed. The planned activities under this grant are for samples approved for batches 3 and 4. Metallic alloy liquids processed in a containerless environment are investigated in all of the proposed measurements for		

Task Description:	QUASI, THERMOPROP and ICOPROSOL. The primary goals of the flight and ground-based experiments are: 1. an investigation of the role of stirring and diffusion on crystal nucleation and a comparison of the experimental results with predictions of a new model for nucleation, the Coupled Flux Model; 2. measurements of key parameters needed for a quantitative analysis of the nucleation data obtained in the ground-based and ISS studies (driving free energy, specific heat and viscosity); 3. measurements of the crystal growth velocity in the ground-based and ISS studies; 4. measurements of the electrical resistivity of metallic liquids as a function of temperature to investigate a proposed correlation between structural changes in the liquid and dynamical properties, and to confirm the existence of a plateau in the temperature dependence of the resistivity that we have recently discovered at high temperatures. Rationale for the Research Measurements of the maximum supercooling of free-cooling metallic liquids that crystallize to phases of a different composition than the liquid, are made in the quiescent environment of the electrostatic levitation facility (WU-BESL) located in our laboratory at Washington University in St. Louis. Long-range diffusion is required for nucleation in these cases. These ground-based results are compared with undercooling studies made as a function of stirring in the ISS-EML. According to the Coupled Flux Model for nucleation, developed by Kelton, the nucleation rate should increase with the stirring rate, since long-range diffusion will become less important. A key goal of our ongoing research is to investigate this. Measurements of the thermophysical properties, including the specific heat, density, surface tension, viscosity and electrical resistivity, will be made using the WU-BESL and the ISS-EML, since these facilities have different capabilities for these studies. The specific heat, density, surface tension and viscosity are needed for a quantitative analysis of the nucleation data. Since our earlier studies have demonstrated that ordering in the liquid couples to the nucleation barrier, the undercooling and thermophysical property measurements will be correlated with x-ray synchrotron diffraction results for the supercooled liquids of interest, obtained previously using WU-BESL. Based on our previous studies, the specific heat and electrical resistivity also provide structural evidence for a cooperative cross over behavior in the shear viscosity at high temperatures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Product advancement in industries such as information technology, aerospace, automotive, and basic and new materials manufacturing requires advanced materials development. In the past, this was often carried out through a laborious process of guided trial and error. The vast increases in computing power have now opened the door for realistic modeling approaches, which have the potential to greatly improve the efficiency of materials development. In the US, this has become a central focus of research in materials science. However, progress is frequently hampered by the lack of relevant thermophysical property data needed for the modeling. Further, key processes such as crystal nucleation from the liquid or other phase transitions within the liquid, are often poorly understood, making it difficult to realistically model them. In many cases, the experimental data and a deeper understanding of some processes, cannot be obtained from terrestrial experiments due to interactions with sample containers at the temperatures of interest. The combination of the containerless based methods of WU-BESL and the ISS-EML enables these measurements. In addition to their relevance for processes such as additive manufacturing on earth, where nucleation in highly stirred conditions is important, measurements in two different gravitational fields will be important for designing additive manufacturing in space and on the Moon. Finally, past measurements of the electrical resistivity in high temperature liquids have revealed a plateau in the temperature dependence of the resistivity at temperatures near a crossover temperature for the viscosity. This is not predicted by any existing theory. Our measurements will provide valuable information for developing a theoretical understanding of this phenomenon.
Task Progress:	Since the new grant began in August 2021, we have made the following accomplishments. 1. Measured and analyzed the effects of stirring on the amount of supercooling in Zr80Pt20, a eutectic forming liquid using the International Space Station (ISS) Electromagnetic Levitator (EML). When the nucleation of the crystal phase requires long-range diffusion, the Coupled Flux Model developed by the Principal Investigator (PI) predicts that the nucleation rate should increase with increased stirring. This was found in ISS measurements for the Vit. 106 liquid, with support from the previous grant. [Ed. Note: This project is a renewal of "NASA Research under ESA-Based Investigations THERMOLAB and ICOPROSOL," grant number NNX16AB52G, with the same PI, Dr. Kenneth Kelton.] For Zr80Pt20 there should be less of an effect of stirring, since only diffusion near the interface of the nucleating eutectic is important. Our results are consistent with this, showing that the nucleation rates differ by a factor of only 1.6 between the weakly stirred liquid and the strongly stirred liquid. Surprisingly, however, the change was a decrease rather than an increase. Our X-ray data have shown that the primary nucleating phase mixture is dominated by a metastable icosahedral phase. The decrease in the nucleation rate with stirring, therefore, could indicate that the stirring disrupted the icosahedral ordering of the liquid. A second possibility, however, is that the stirring disrupted the local diffusion fields at the interface of the nucleating eutectic phase mixture. Further study is needed to investigate this. 2. Measured the specific heat, Cp, as a function of temperature in Zr80Pt20 and Pt57.5Cu14.7Ni5.3P22.5 liquids with the ISS-EML. The step modulation calorimetry data showed an increase in Cp for temperatures lower than the dynamical crossover temperature (TA), where the viscosity changes from an Arrhenius temperature dependence to a super Arrhenius dependence, signaling an increase in the activation energy with decreasing temperature. Molecular dynamics studies indicate that this change signals the onset of cooperativity in the liquid dynamics. Our results agree with this and further show the predicted link between dynamics and structure in liquids. 3. Measured the electrical resistivity of liquids Zr80Pt20 and Pt57.5Cu14.7Ni5.3P22.5 using the ISS-EML. Below TA, the resistivity begins to rise, again giving evidence for a link between the dynamics and structures of metallic liquids. They also show that at higher temperatures, the resistivity is independent of the temperature. This is in agreement with our earlier measurements of the resistivity of metallic liquids and suggests that the saturation of resistivity at high temperatures is likely a generic behavior in many metallic liquids. A quantitative explanation of this phenomenon does not yet exist. A qualitative explanation is that the dynamic disorder in the liquid changes so fast above TA that the electrons are unable to track the changing local structure of the liquid, which appears to be static for the electrons. Moreover, scattering by high frequency phonons also becomes ineffective, since the atoms cannot communicate with each other as the structural changes become faster than the time it takes for a phonon to travel the distance between the

	atoms. Both of these effects may make electron scattering mechanisms ineffective, resulting in a saturation of the electrical resistivity. However, it is important to emphasize that this behavior is not predicted by any existing theory of the electrical resistivity of metallic liquids and requires a new theoretical treatment. 4. We have developed the parameters for two batch 3 samples, Zr47Cu47Al6 and Ti45Zr45Ni10, with representatives of the Microgravity User Support Center (MUSC), and two European Space Agency sponsored projects: "Thermophysical properties and solidification behaviour of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order" (ICOPROSOL), and "Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products" (THERMOPROP). 5. We have prepared and delivered samples of Zr50Pd50 and Ti40Zr27Ni33 and the necessary documentation to MUSC for electromagnetic coupling constant measurements. 6. We have made thermal evaporation loss measurements for two batch 4 samples, Zr50Pd50 and Ti40Zr27Ni33. 7. We have prepared a contribution to a joint publication entitled, “Microgravity Containerless Processing – Rapid Solidification", that is being written for an npj Microgravity Special Issue on the SciSpace White Papers - Materials Science. 8. Kelton led the preparation of a topical white paper submitted to the Decadal Survey on Biological and Physical Sciences Research in Space 2023-2032, entitled, “Nucleation Investigation and Control – The Next Decade and Beyond in Space Materials and Soft Matter Research”.
Bibliography Type:	Description: (Last Updated: 05/20/2025)
Books/Book Chapters	Kelton KF. "Theory of nucleation and glass formation." in "Metallurgy in Space: Recent Results from ISS. The Minerals, Metals & Materials Series." Ed. HJ. Fecht, M. Mohr. Cham, Switzerland: Springer International, 2022. p. 153-78. https://doi.org/10.1007/978-3-030-89784-0 , Apr-2022
Books/Book Chapters	Gangopadhyay AK, Kelton KF. "Electrical resistivity measurements on the International Space Station for the studies of dynamics in metallic liquids." in "Metallurgy in Space. The Minerals, Metals & Materials Series." Ed. HJ. Fecht, M. Mohr. Cham, Switzerland: Springer International, 2022. p. 451-69. https://doi.org/10.1007/978-3-030-89784-0_20 , Apr-2022